Project Report

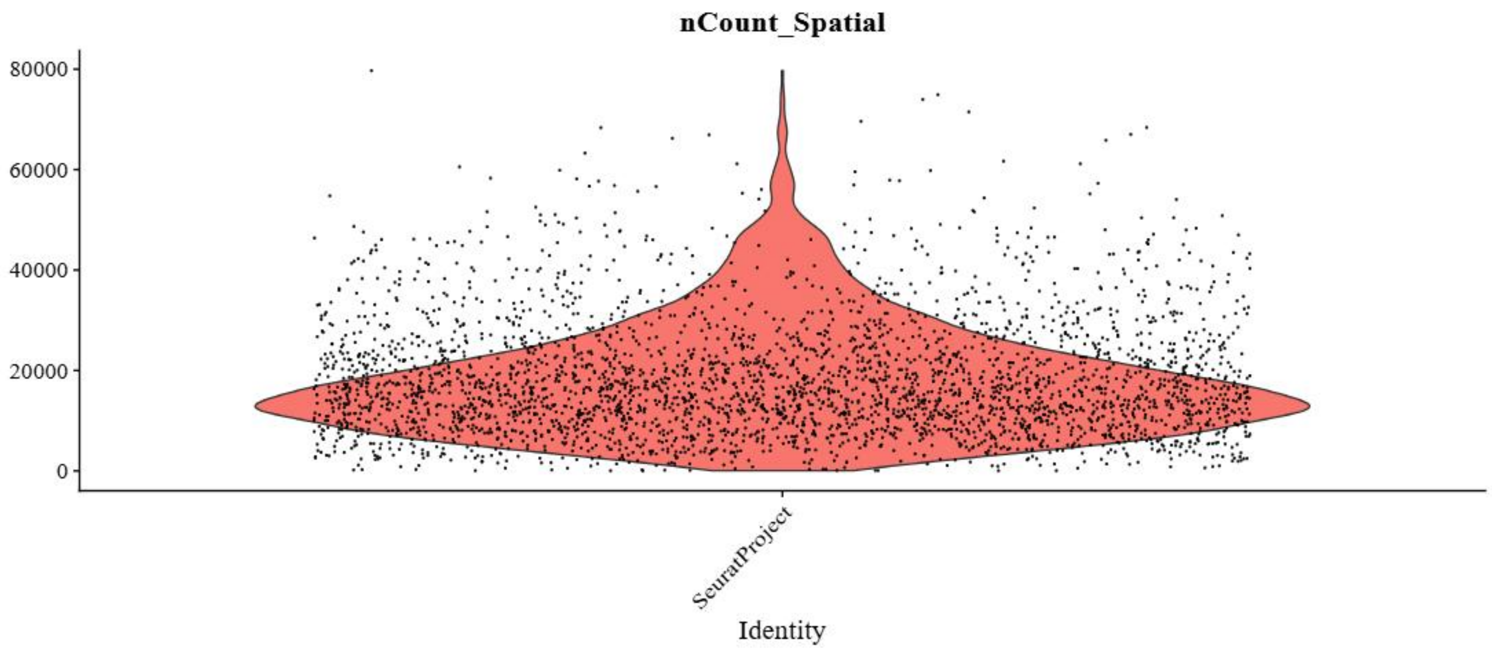
Inspecting the Seurat-object

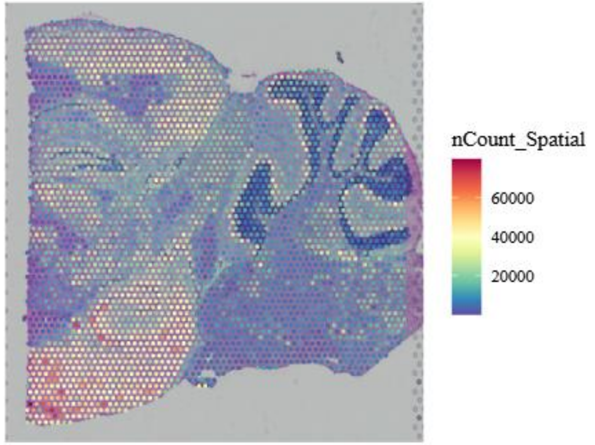
Gene Expression Data stored in the assay named "Spatial“, we can access it through @assay slot **“sample1@assays” command**

Tissue Image stored in the @images slot , we can access it through @images **“sample1@images” command**

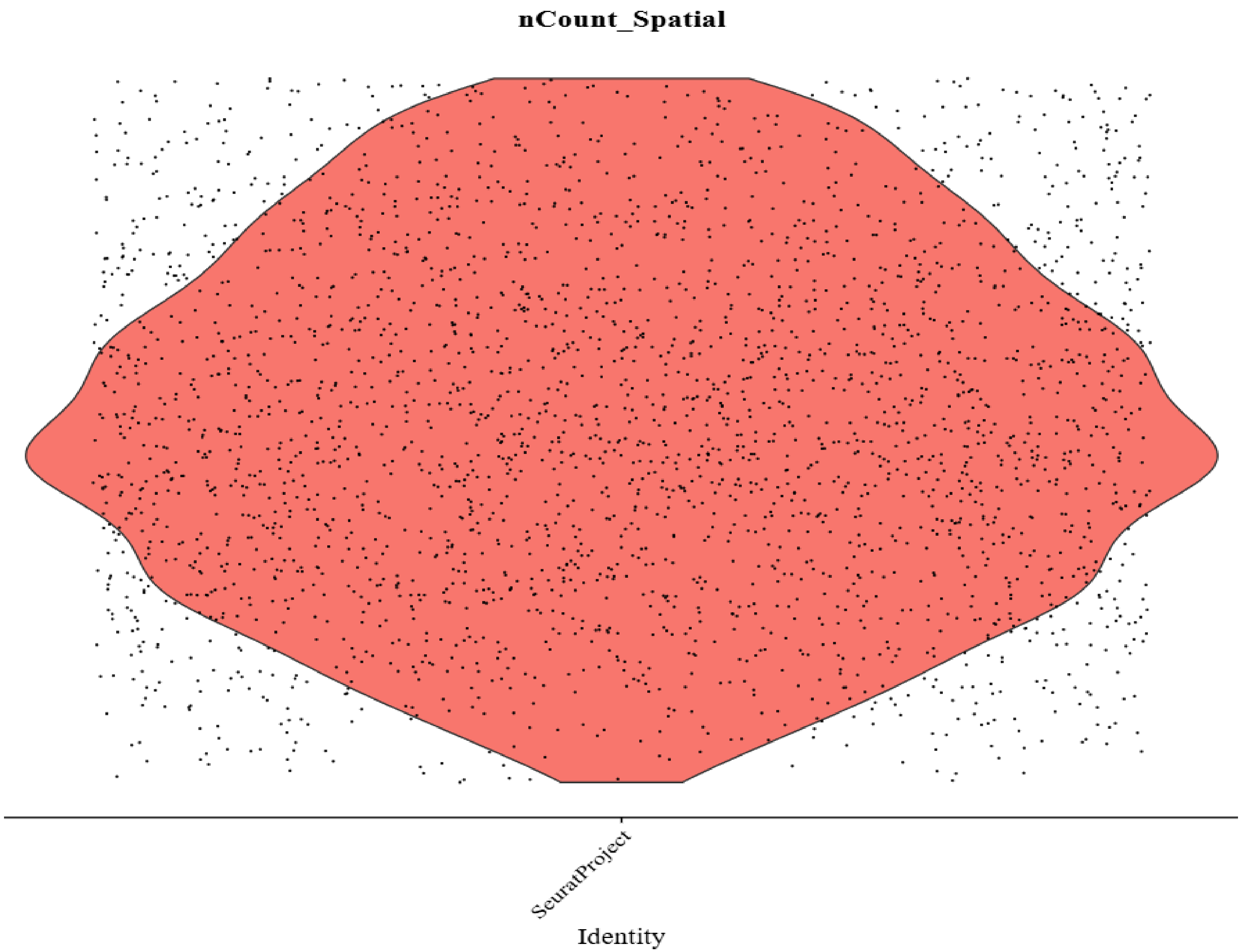
We can access the data from Seurat : **by using “sample1 @ meta . data” command**

Visualization of a feature



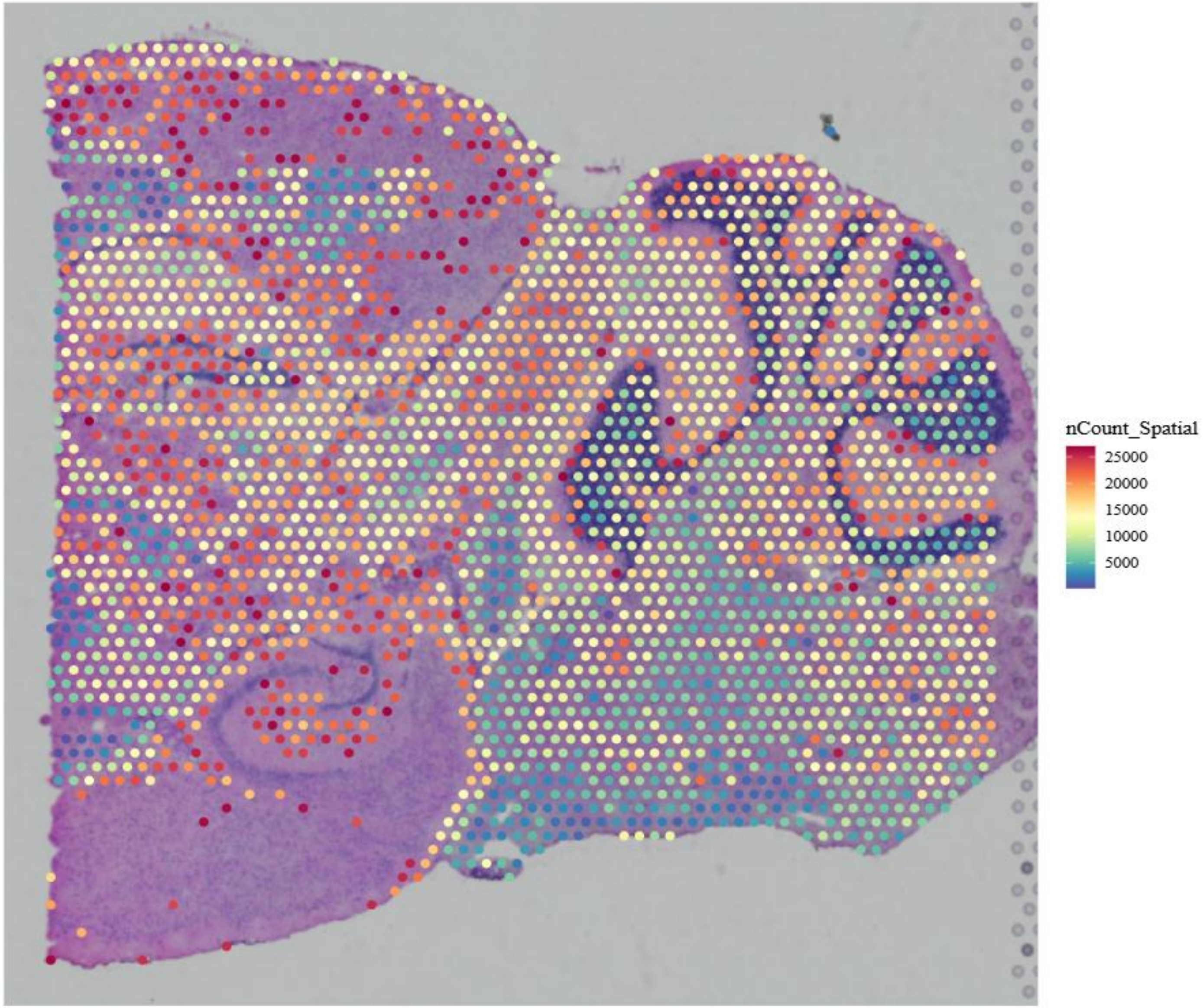


After filtration sample 1

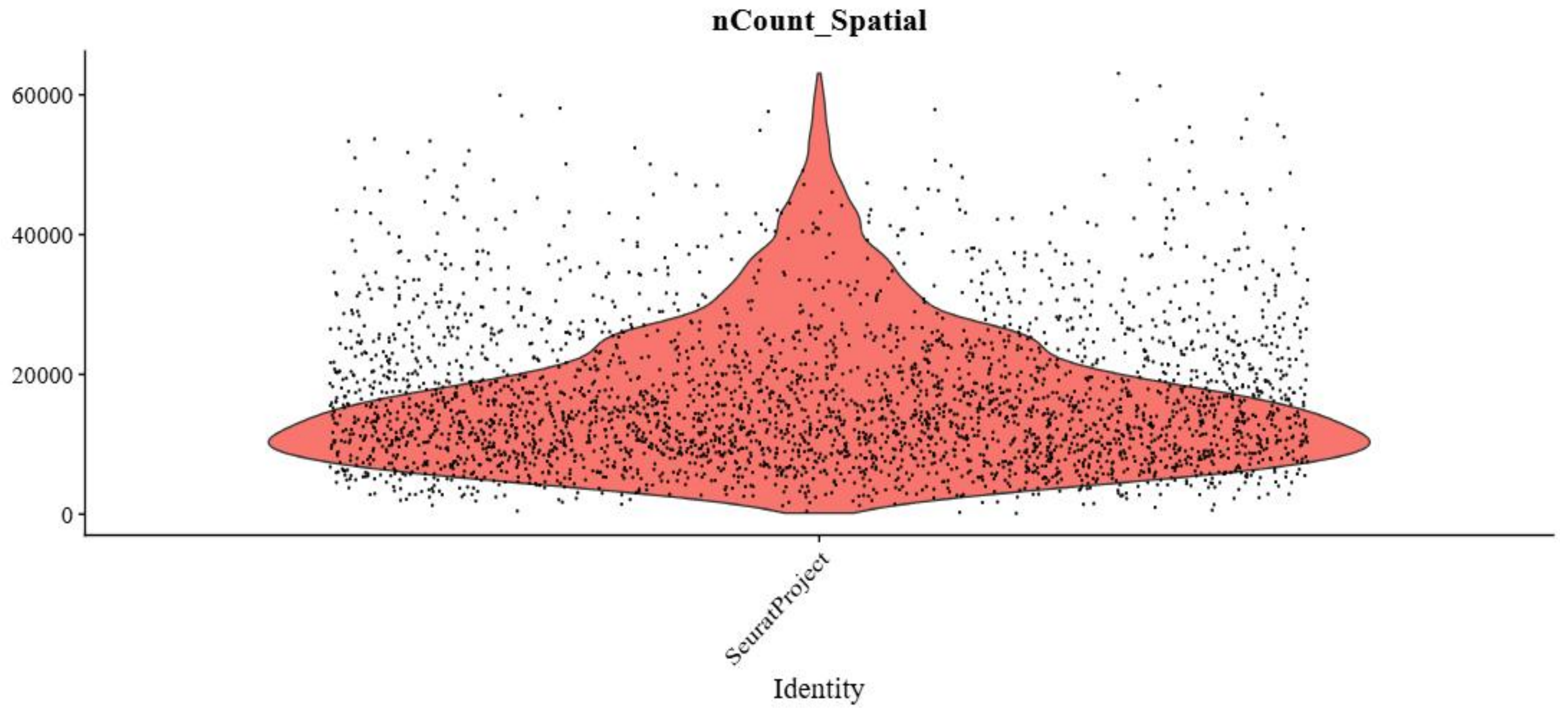


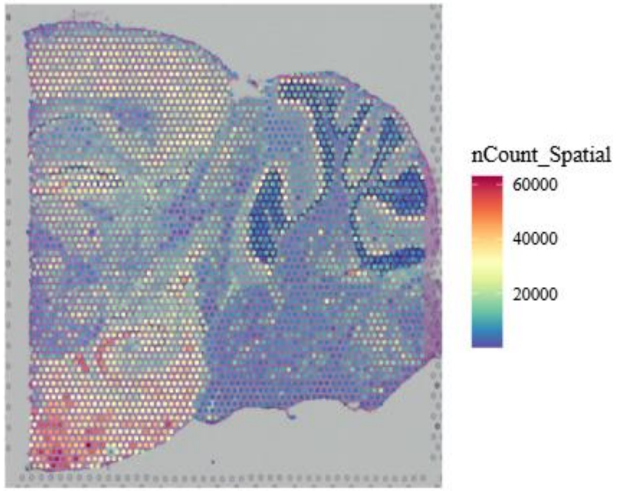
Sample 1

After filtration sample 1

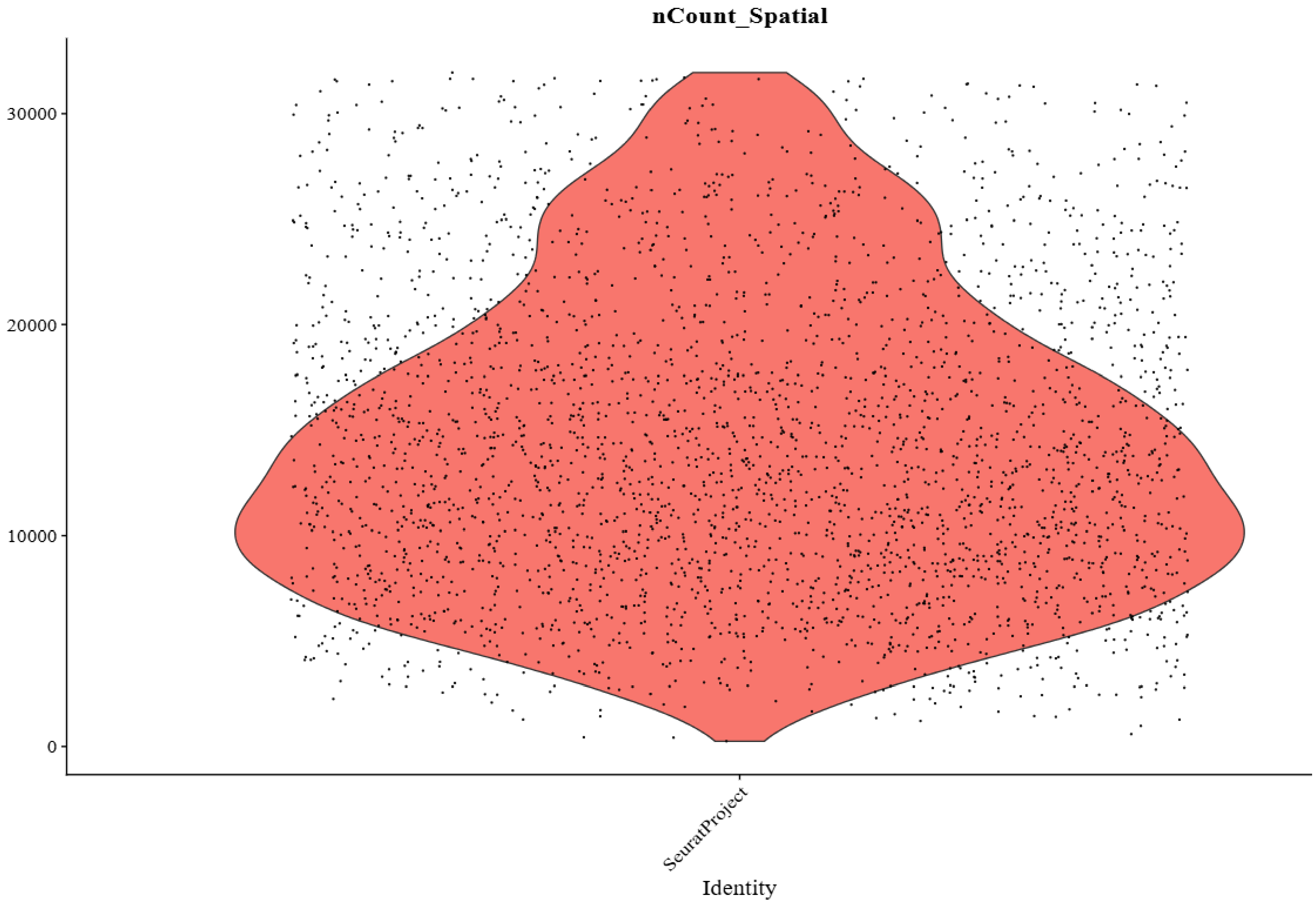


Visualization of a feature

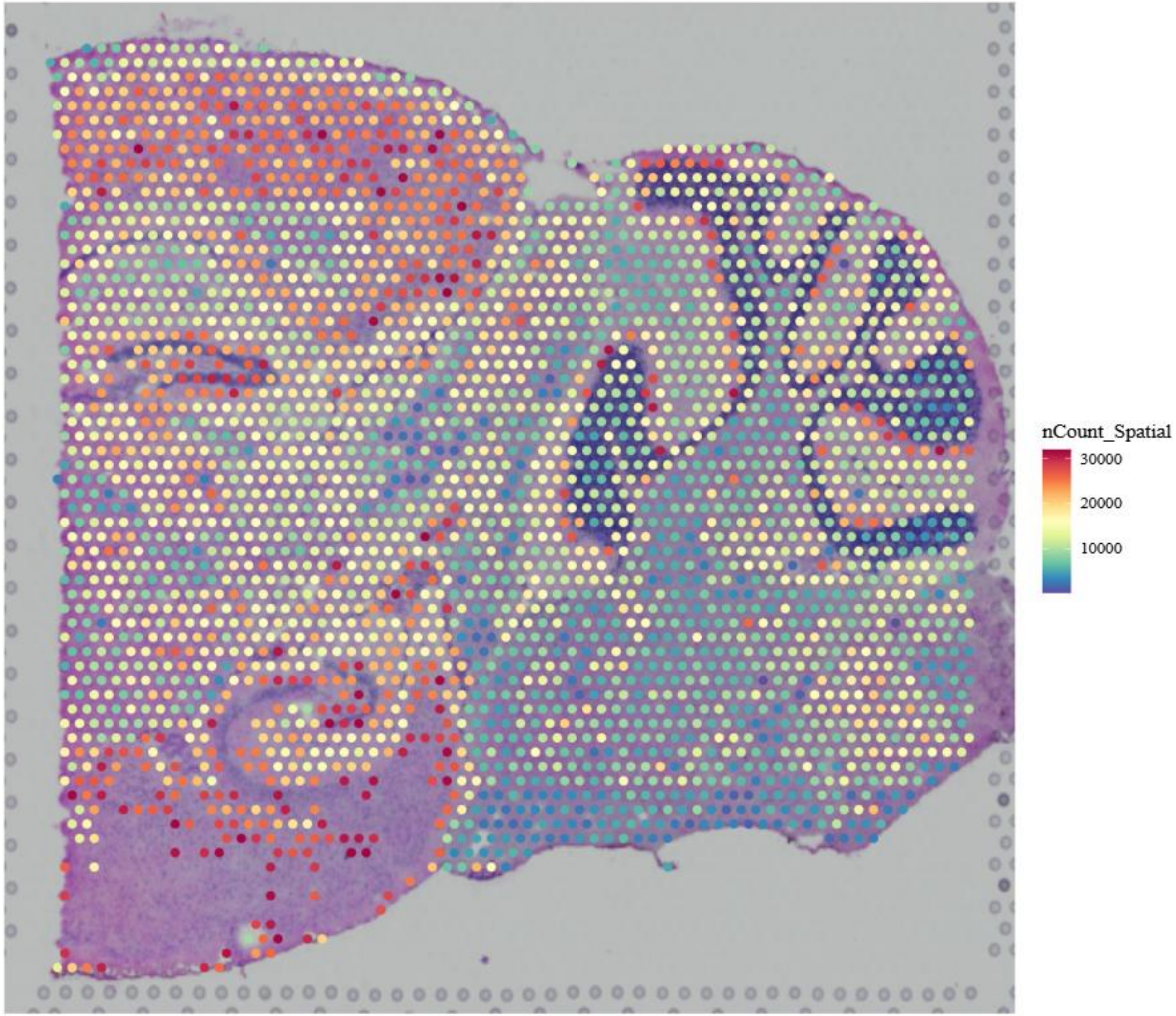




After filtration sample 2



After filtration sample 2



Filtering

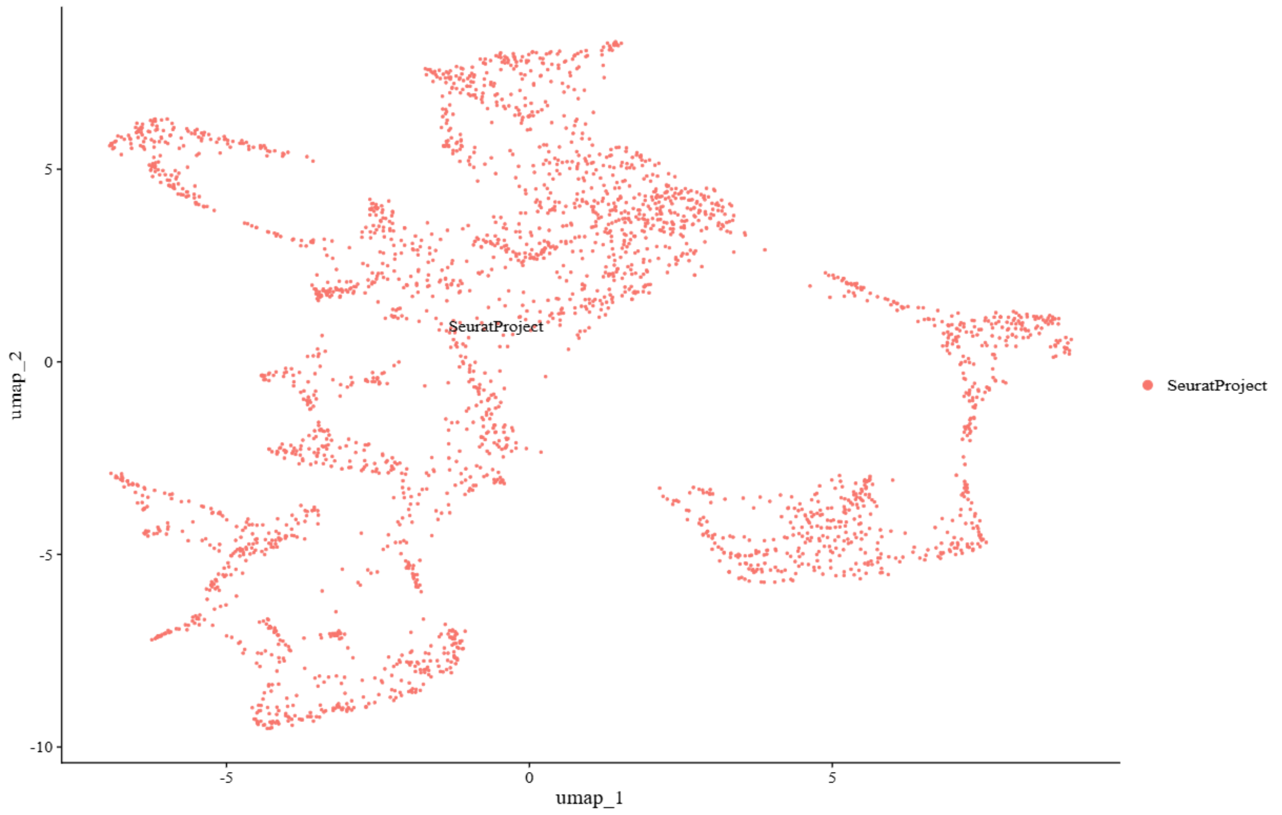
- In scRNA-seq, the filtering range is 200–2500 genes per cell, while in spatial data, it's 200–27,000/32,000 counts per spot because spots capture RNA from multiple cells. Spatial thresholds are higher due to this difference. Violin and spatial plots confirm these thresholds remove noise and artifacts while keeping useful data.

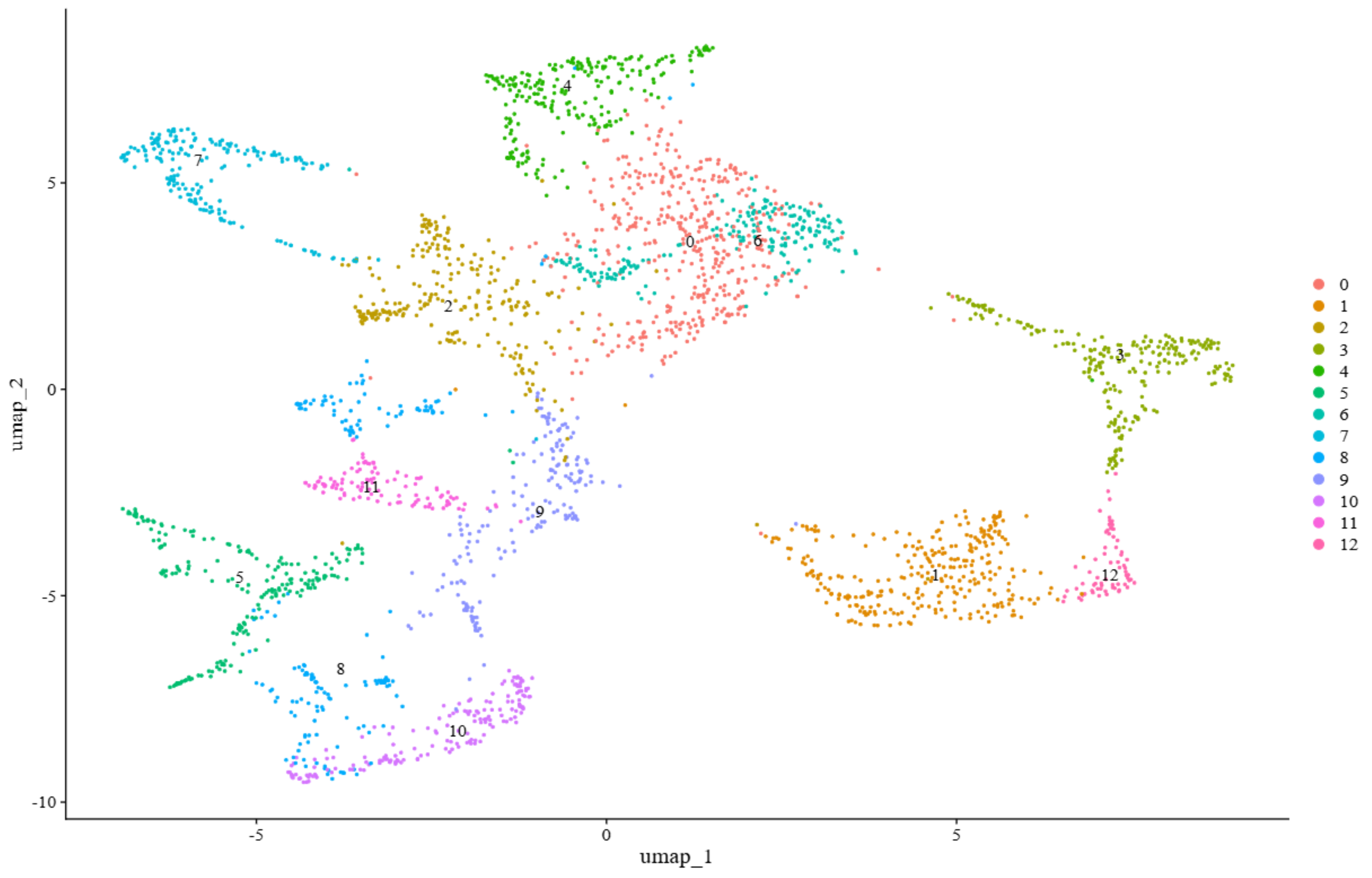
Apply scTransform

normalization steps are replaced by ScTranforme and this method create a STC array.

Sample 1

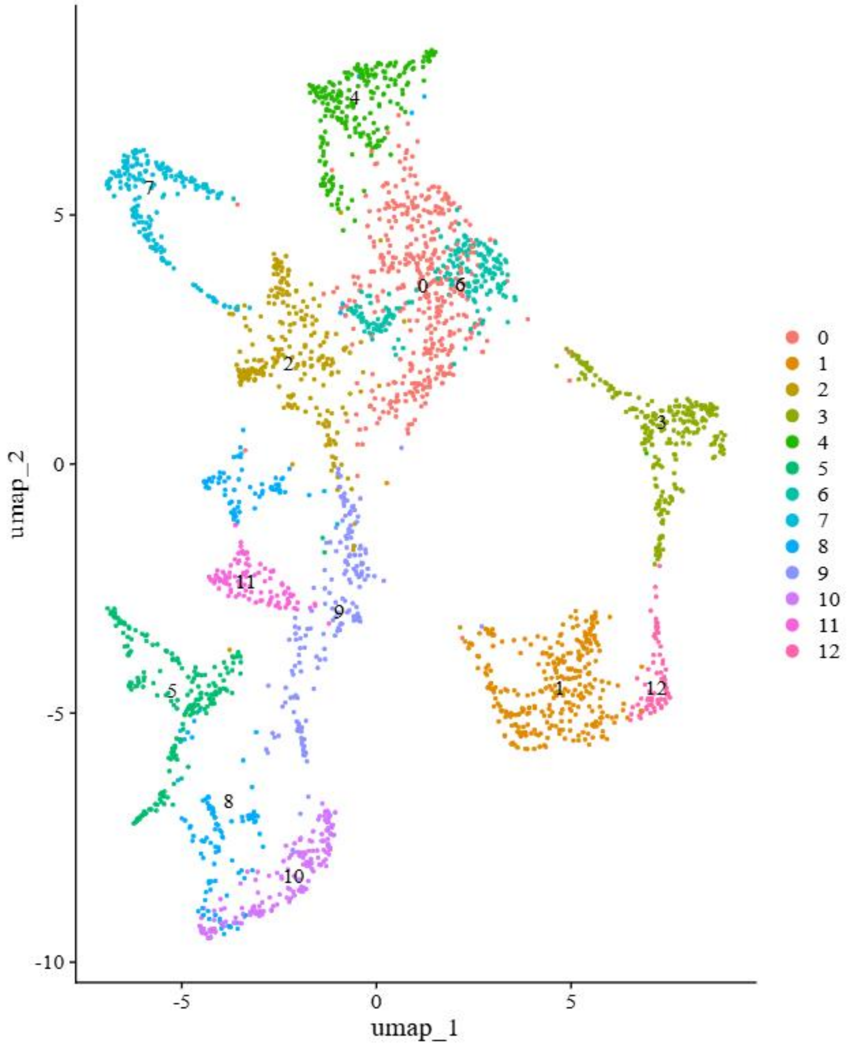
Dimensionality reduction

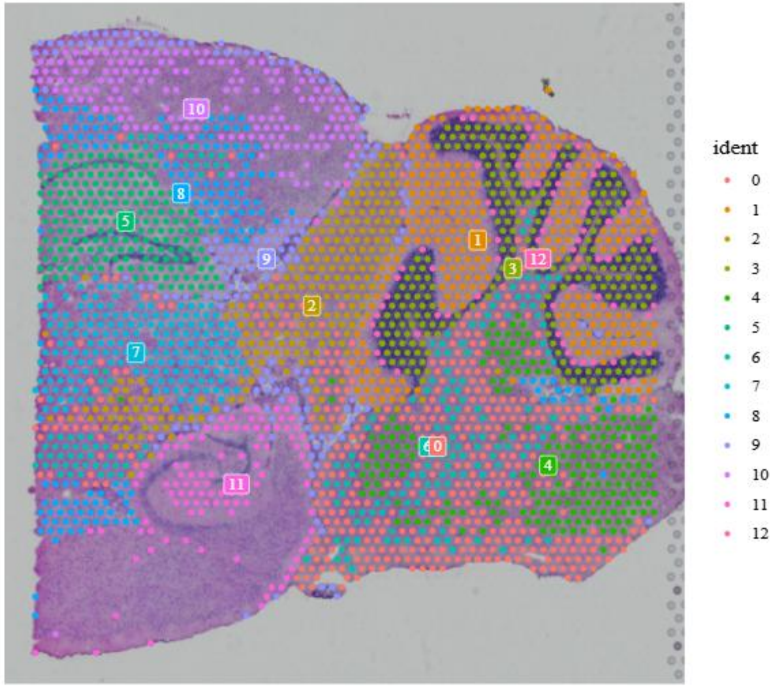




Clustering

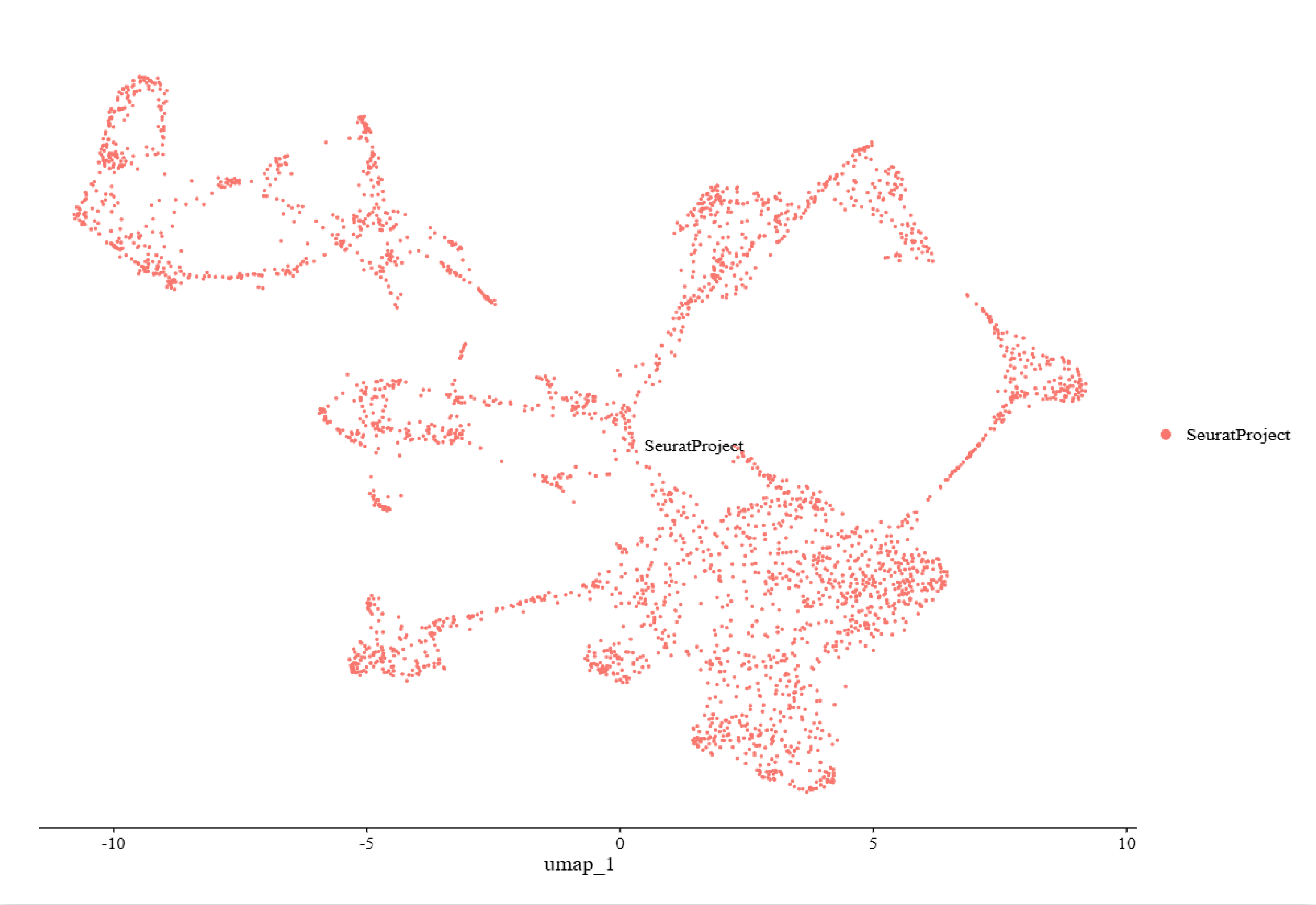
Sample 1





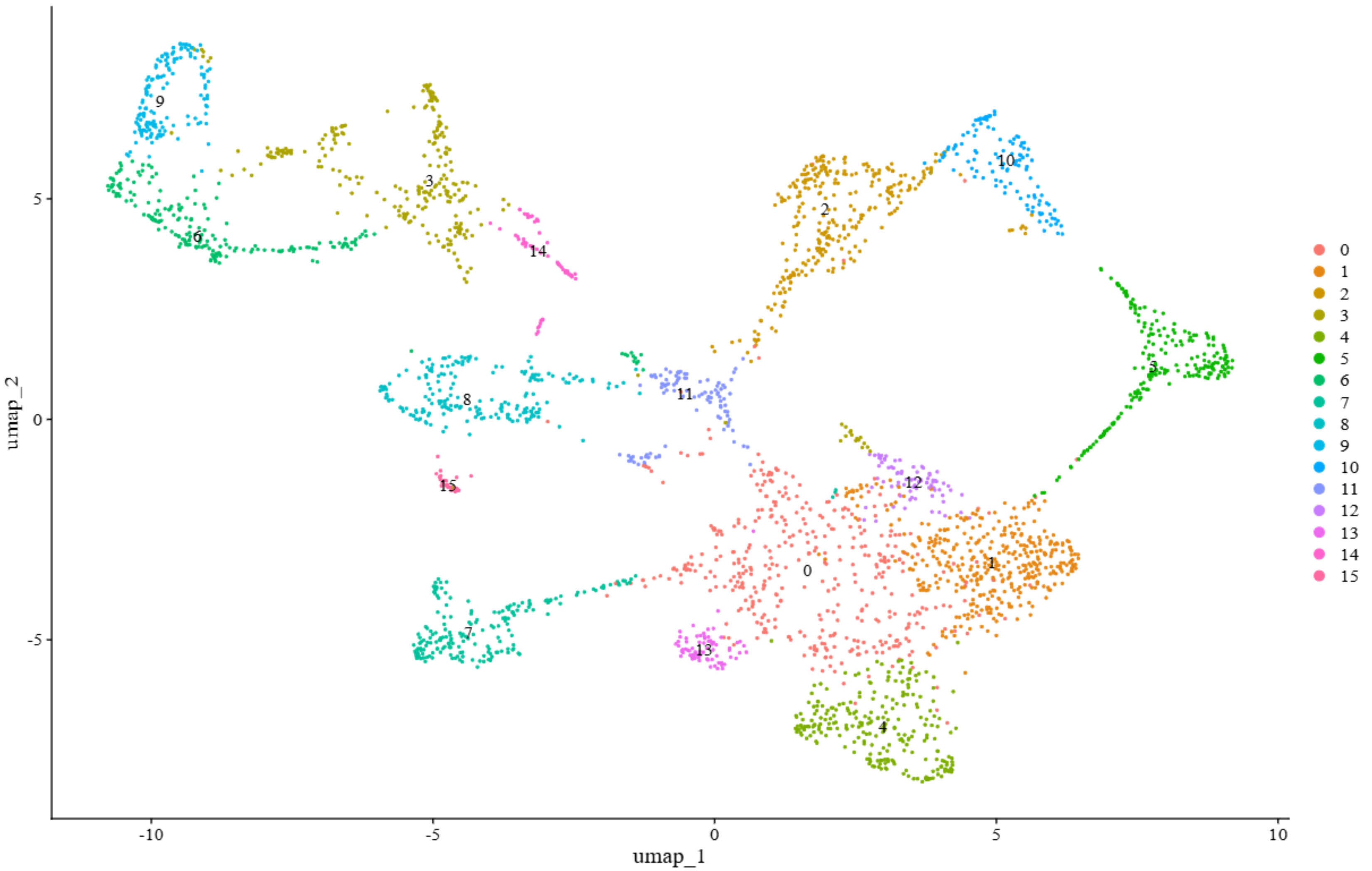
Clustering

Sample 1



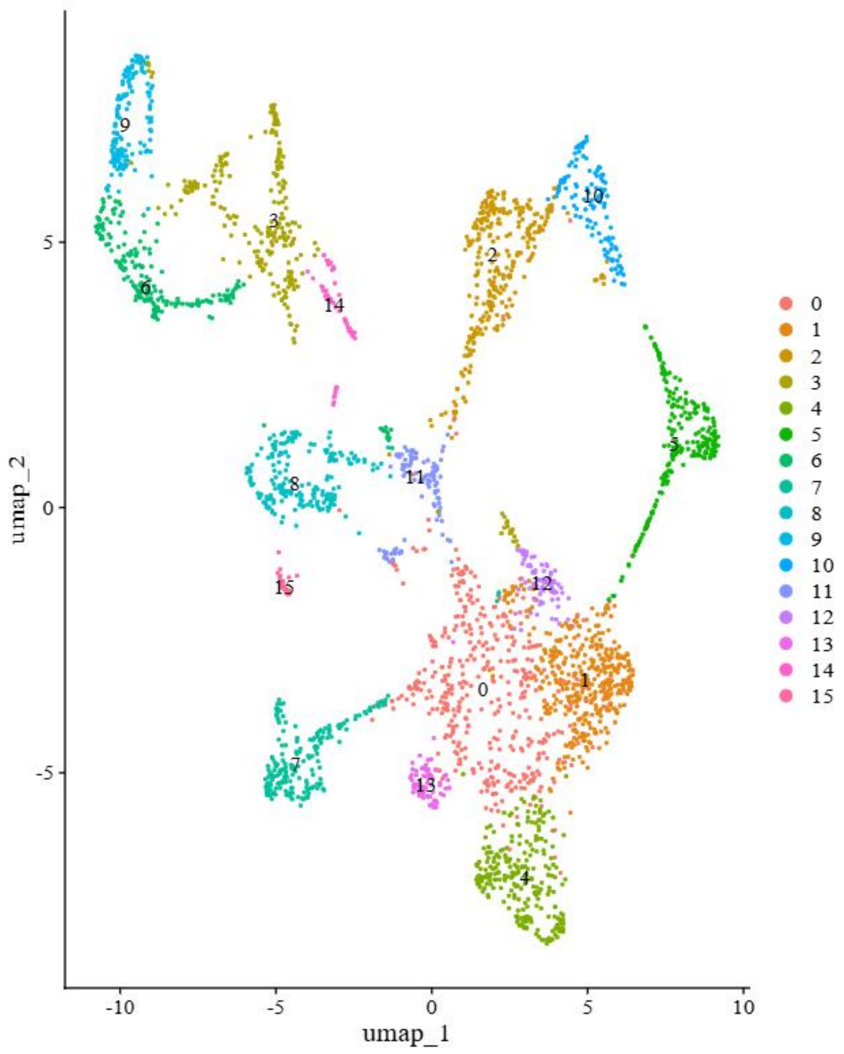
Dimensionality reduction

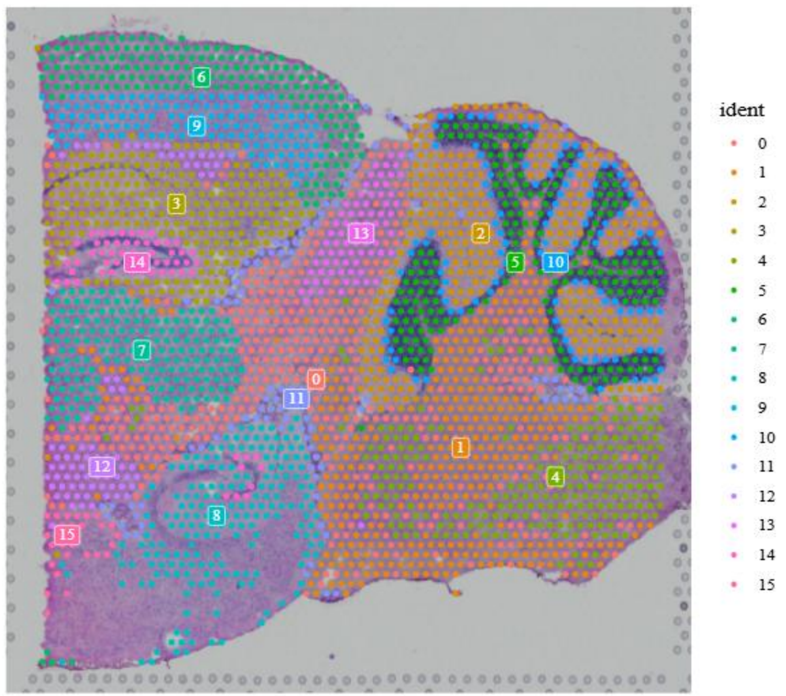
Sample 2



Clustering

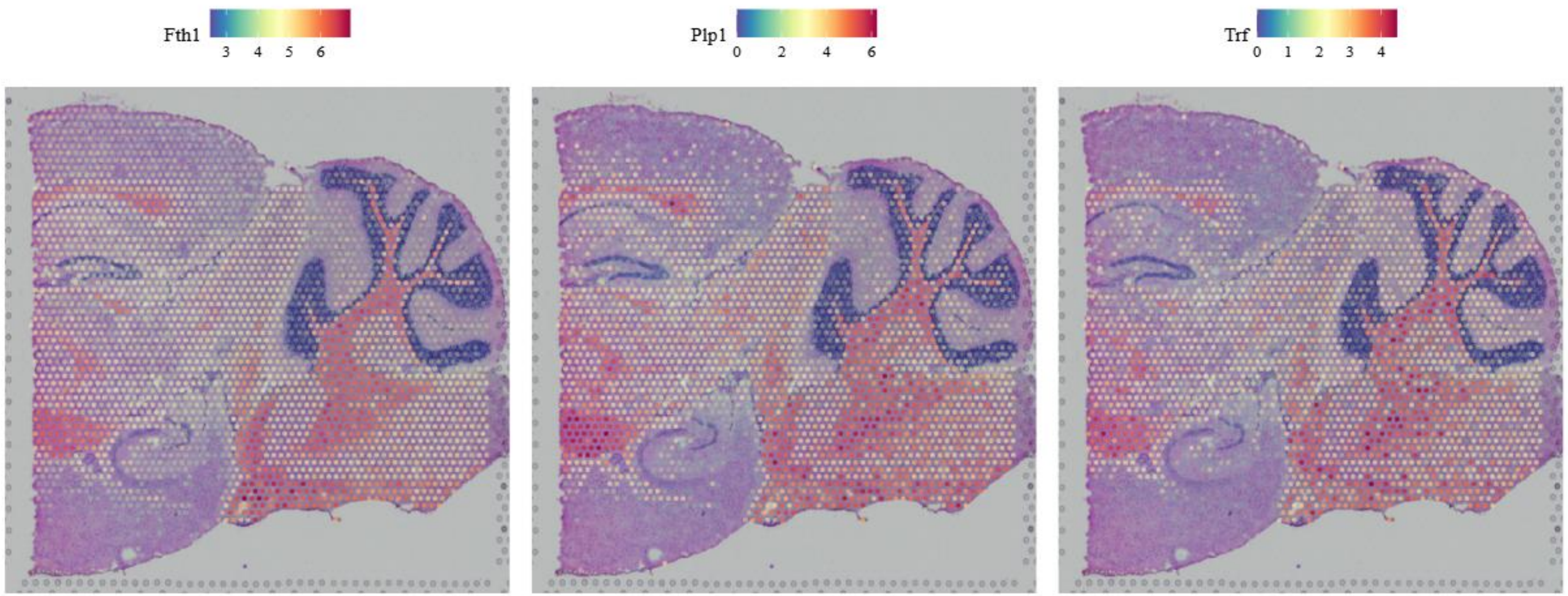
Sample 2



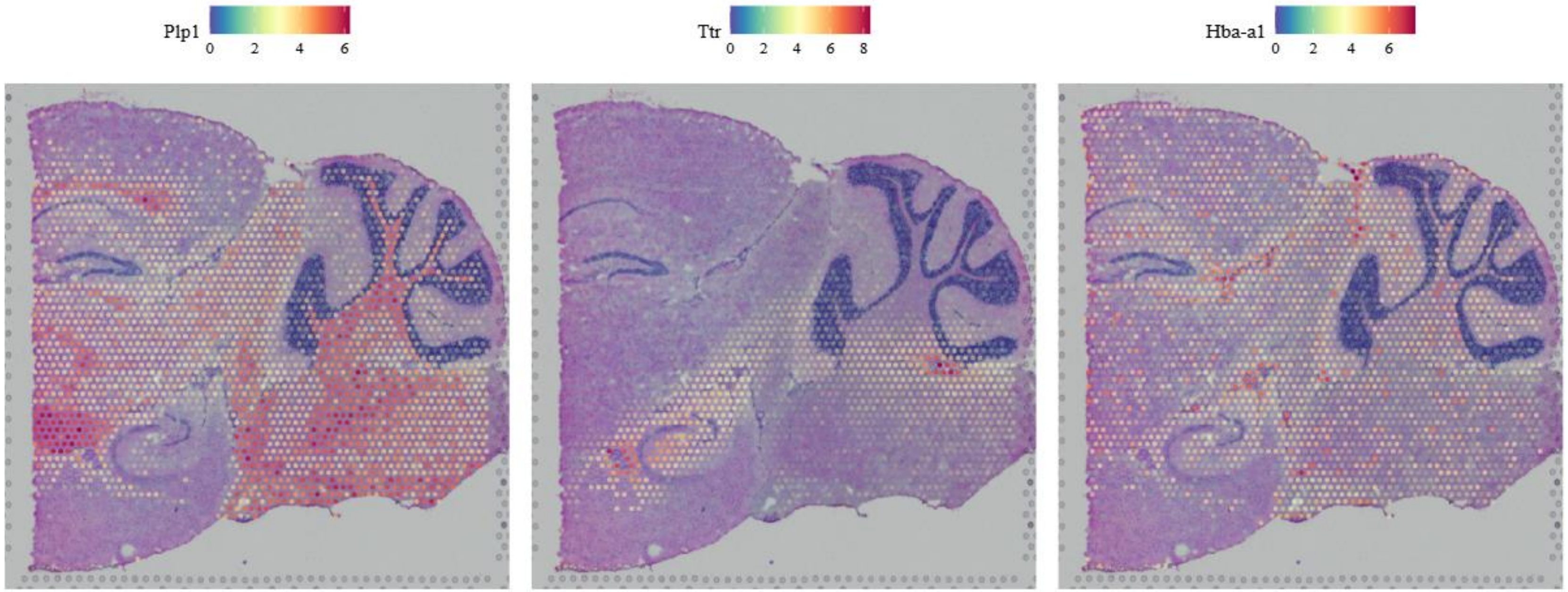


Clustering

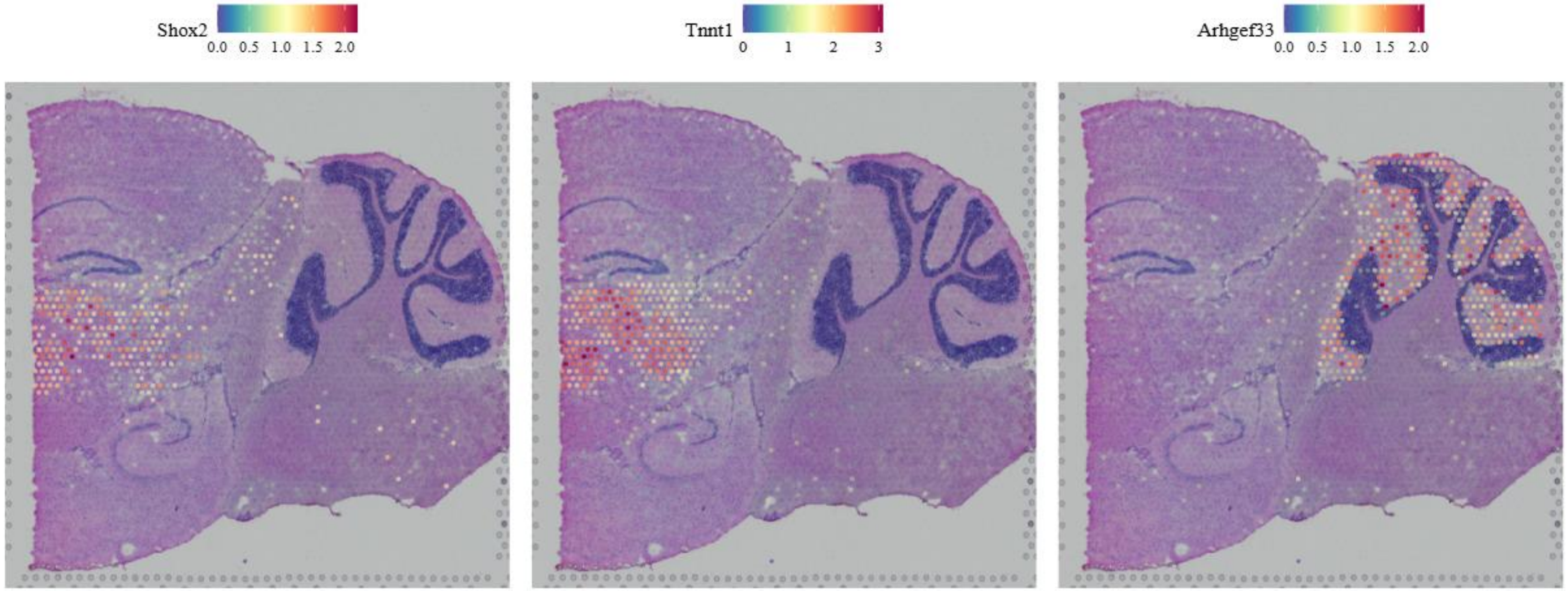
Sample 2



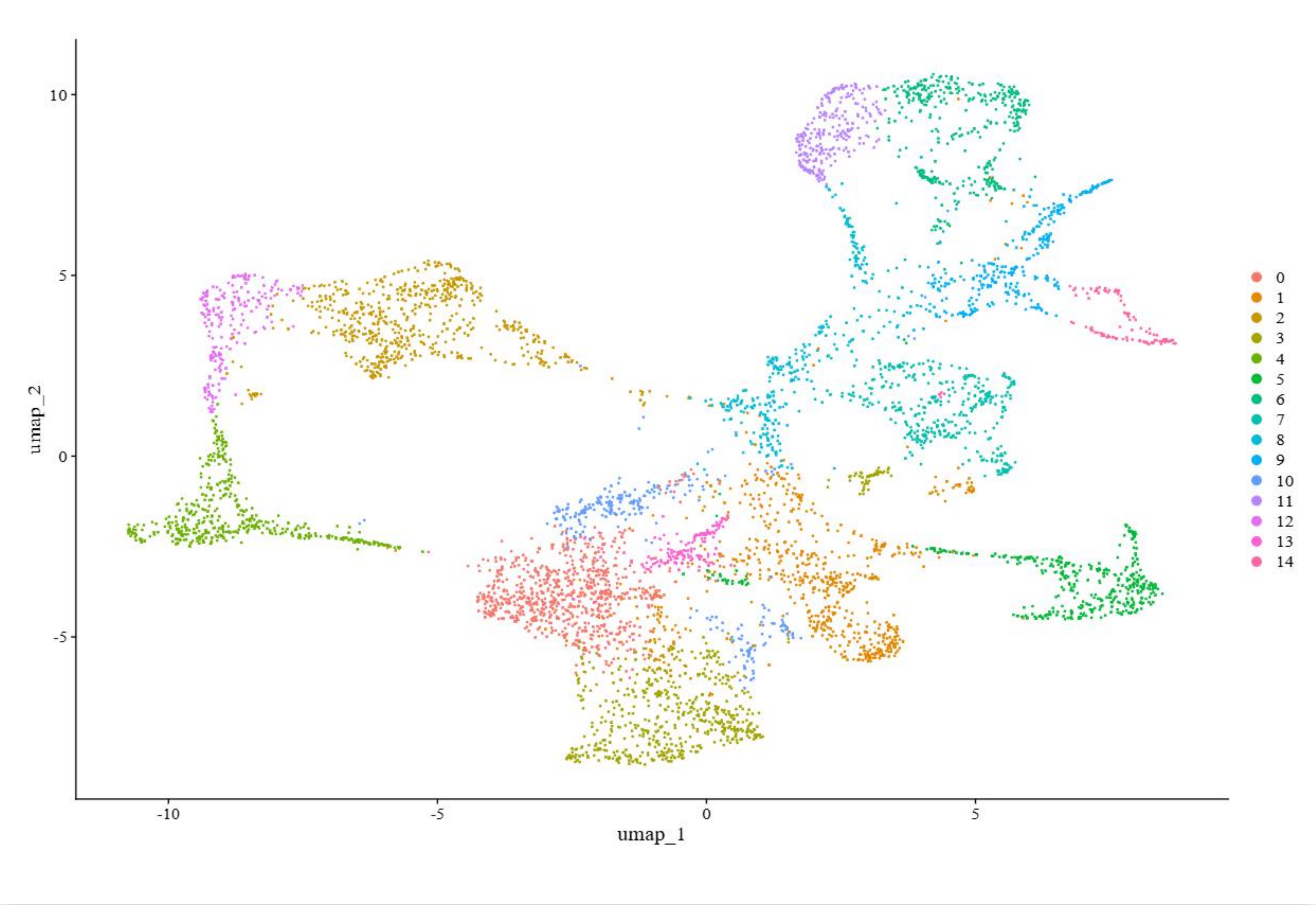
DEG (Differentially Expressed Genes) analysis based on the clustering



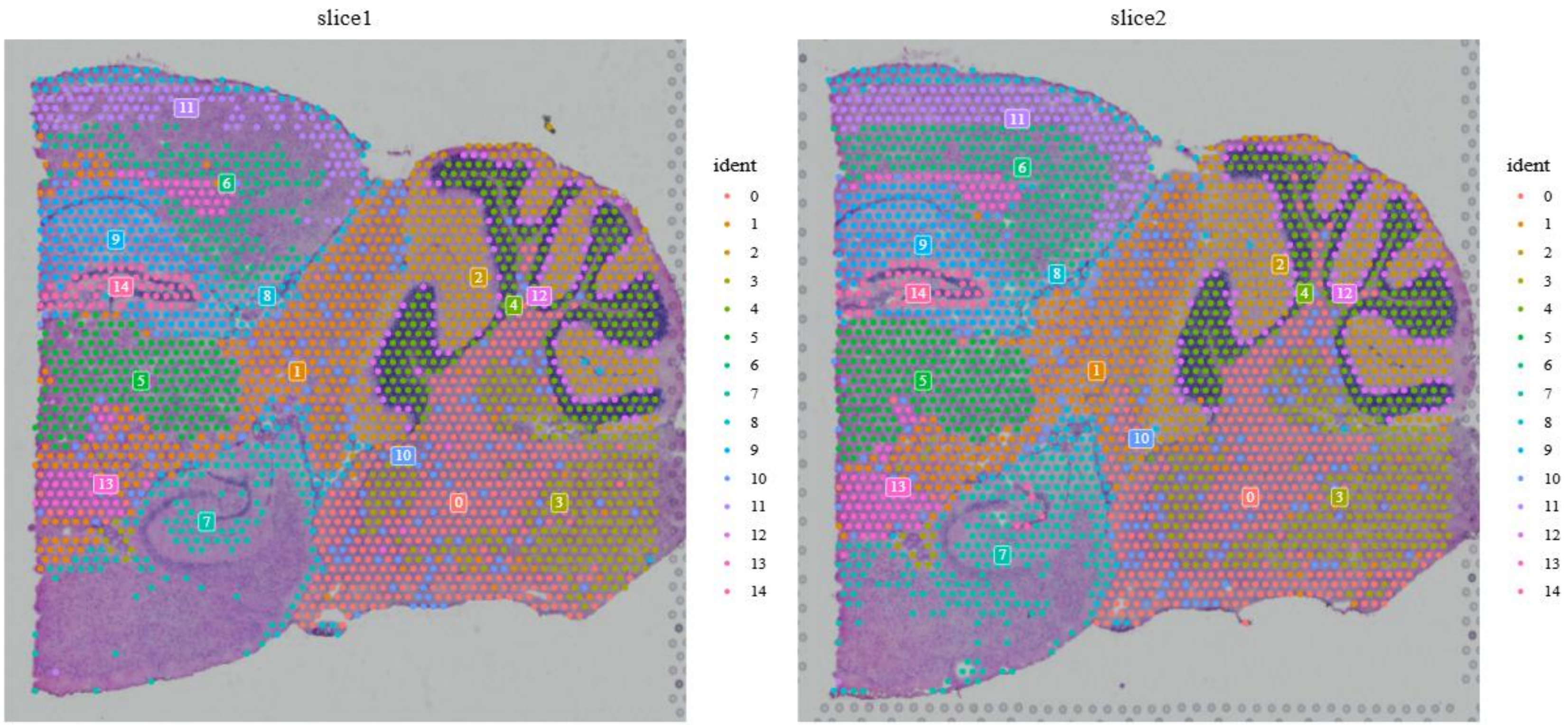
DEG (Differentially Expressed Genes)based on spatial patterning



DEG (Differentially Expressed Genes)based on p-value



Merging without Batch-correction



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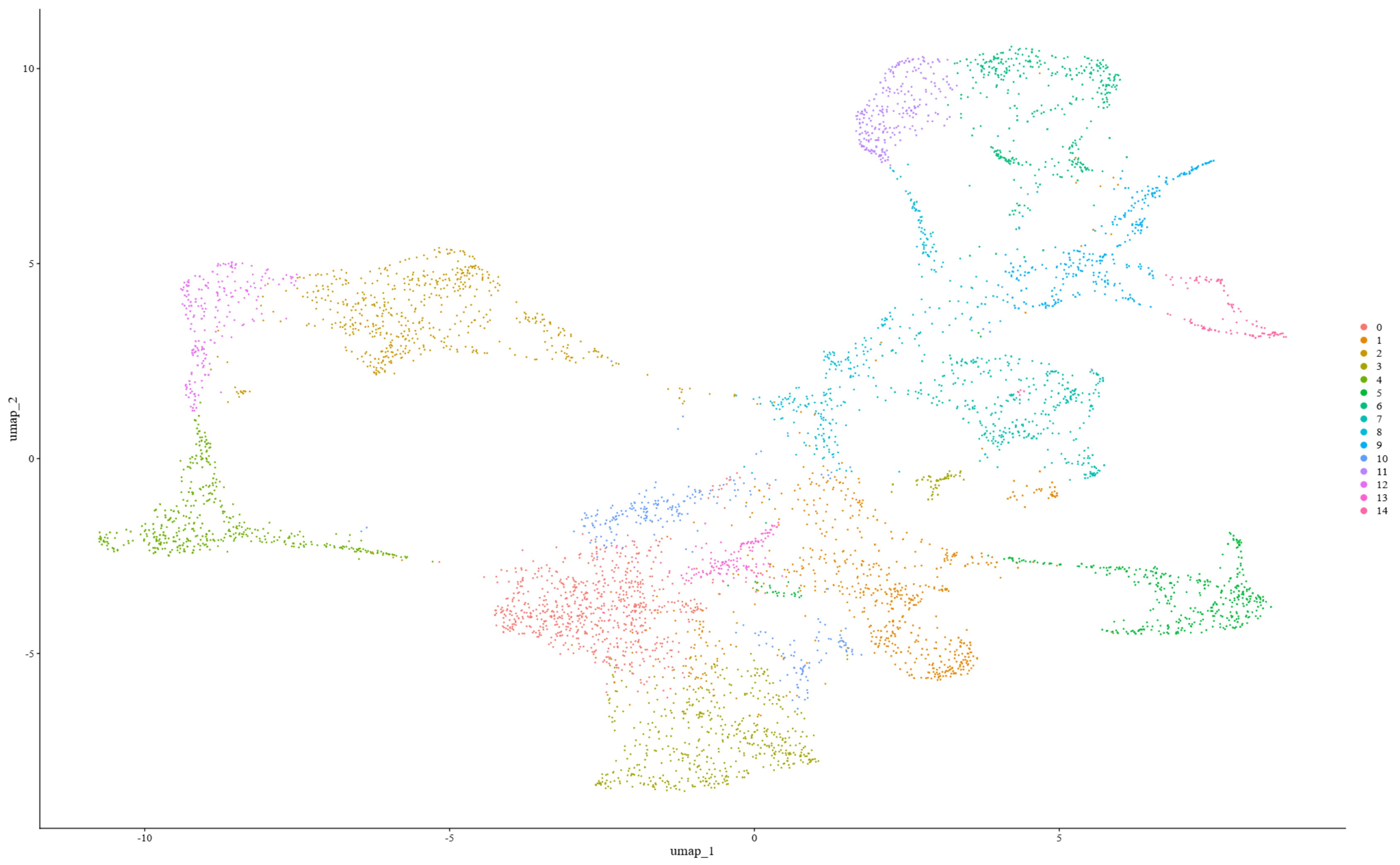
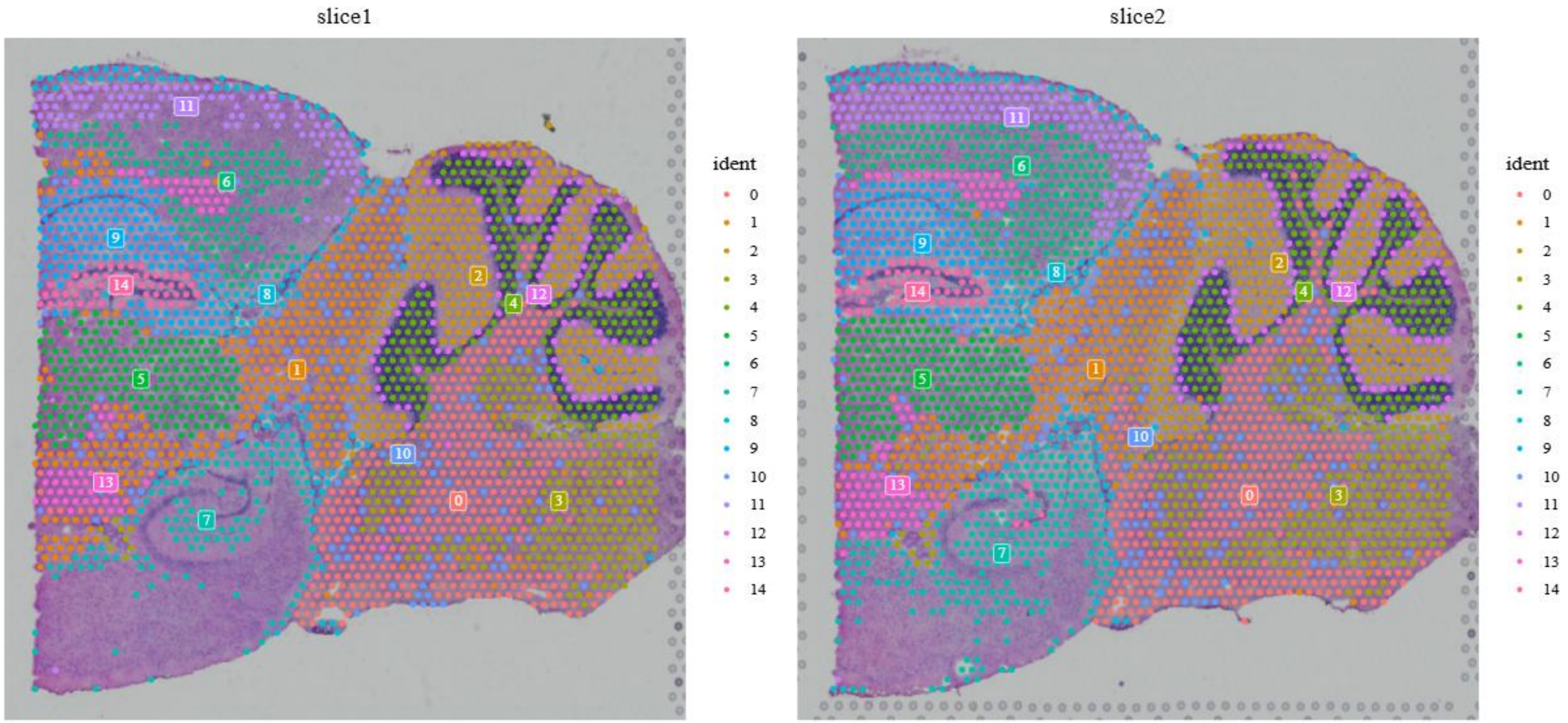
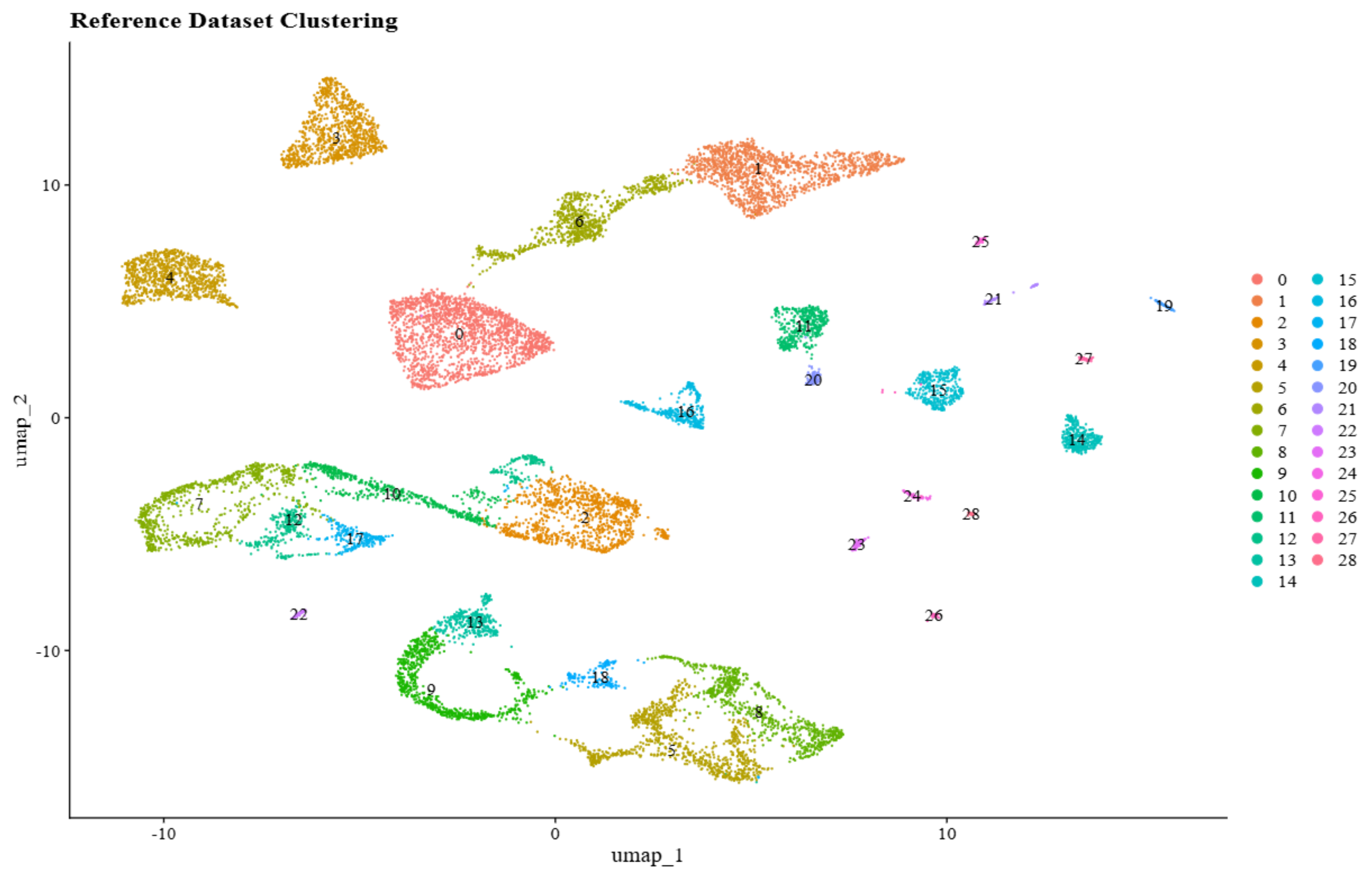


Merging with Batch-correction

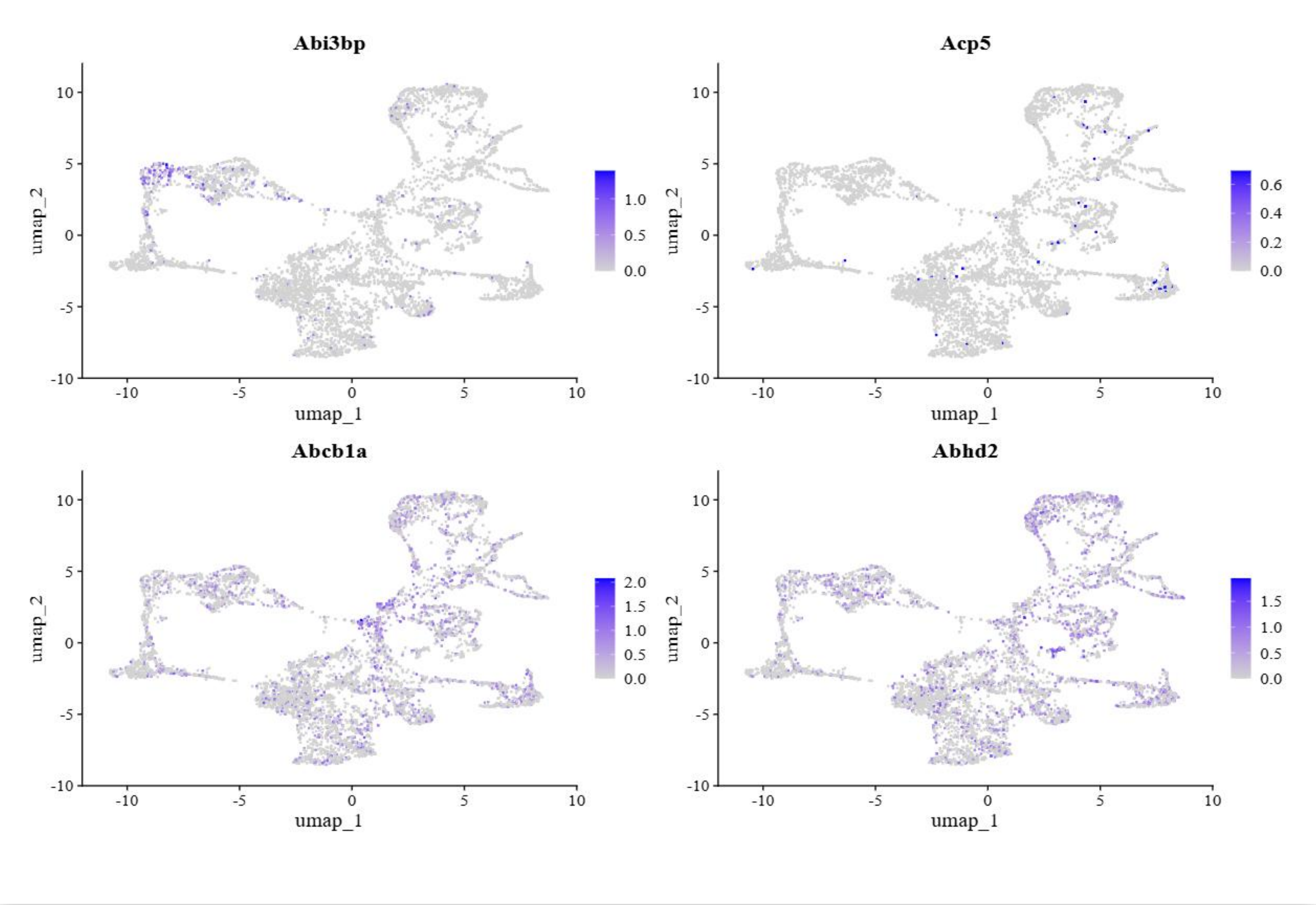


Merging with Batch-correction

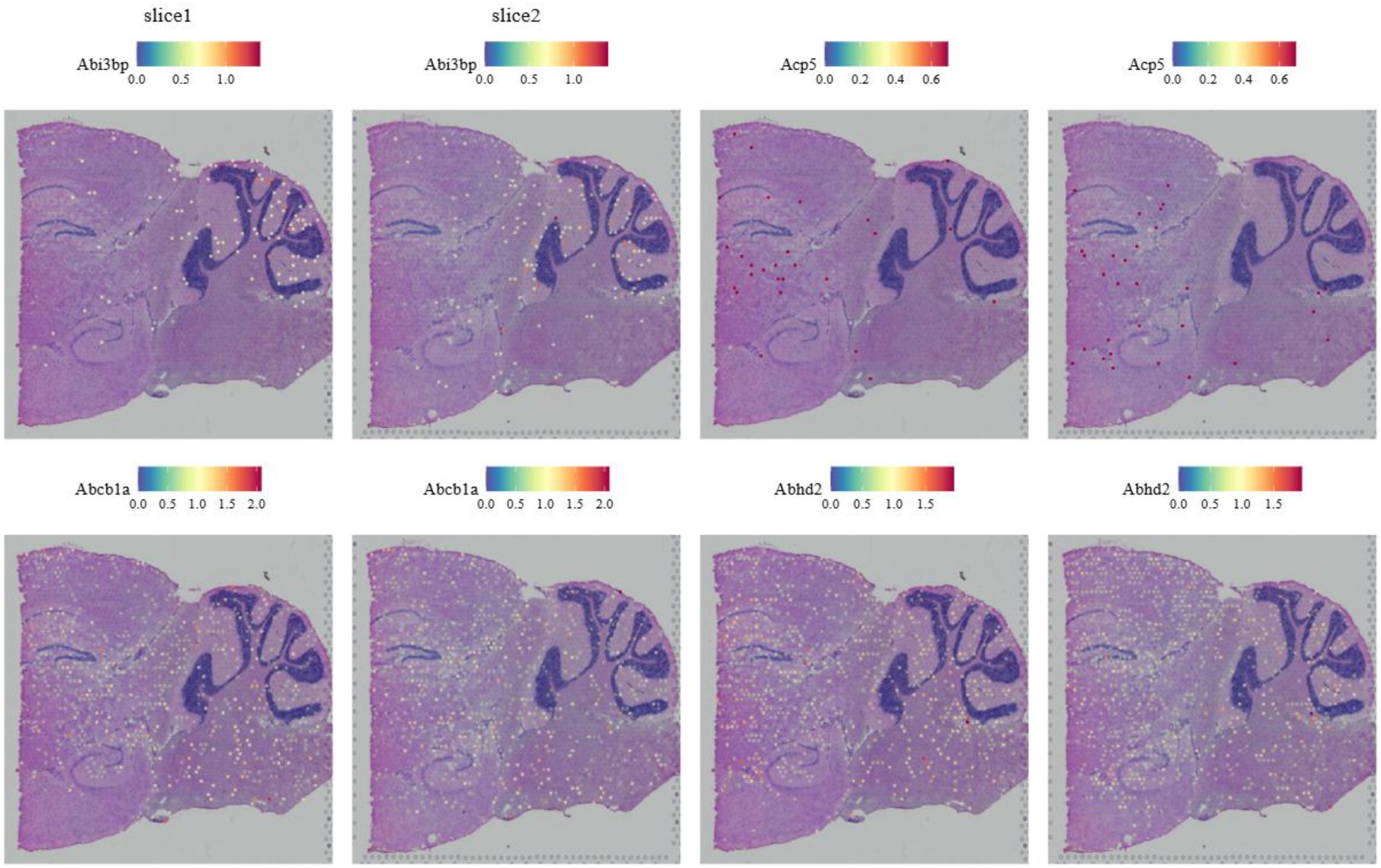
There is no need for batch correction while The clusters were well-distributed and synchronized after performing the bacth correction.



Automatic Annotation using Data Integration



Manual Annotation, Marker Gene Expression in UMAP



Manual Annotation, Marker Gene Expression in UMAP on the tissue slides